

Second Partial

Fco. Vazquez

Tec de Monterrey

Rule of derivatives

Power rule

$$\frac{d}{dx}x^n = nx^{n-1}$$

Chain rule

$$\frac{d}{dx}u(v(x)) = u'(v(x)) \cdot v'(x)$$

Quotient rule

$$\frac{d}{dx} \frac{u(x)}{v(x)} = \frac{u'v - uv'}{v^2}$$

Product rule

$$\frac{d}{dx}u(x)v(x) = u'v + uv'$$

Example: $f(x) = e^{-x^2} \sin(2\pi x)$

Product rule

First we identify that is a product of a composition of functions. It is a product, because we can write $f(x) = u(x)v(x)$. Where $u(x) = e^{-x^2}$, and $v(x) = \sin(2\pi x)$.

Chain rule

Then, looking at $u(x)$ and $v(x)$ we can see that is a composition of functions, $u(x) = (e^x \circ -x^2)(x) = e^{-x^2}$ and $v(x) = (\sin(x) \circ 2\pi)(x) = \sin(2\pi x)$.

Example: $f(x) = e^{-x^2} \sin(2\pi x)$

Product rule

Let's start with the product rule,

$$\begin{aligned} f'(x) &= u'v + uv' \\ &= \left(\frac{d}{dx} e^{-x^2} \right) \sin(2\pi x) + e^{-x^2} \left(\frac{d}{dx} \sin(2\pi x) \right) \end{aligned}$$

Now, it is time to use the chain rule to compute the derivatives of u and v .

Example: $f(x) = e^{-x^2} \sin(2\pi x)$

Chain rule: e^{-x^2}

As we identify before, $e^{-x^2} = e^x \circ -x^2$, therefore,

$$\begin{aligned}\frac{d}{dx}e^{-x^2} &= \frac{d}{dx}e^x \Big|_{x=-x^2} \left(\frac{d}{dx} -x^2 \right) \\ &= e^x \Big|_{x=-x^2} (-2x) \\ &= e^{-x^2} (-2x) \\ &= -2xe^{-x^2}\end{aligned}$$

Example: $f(x) = e^{-x^2} \sin(2\pi x)$

Replace

Let's replace the previous result into the solution,

$$\begin{aligned} f'(x) &= \left(\frac{d}{dx} e^{-x^2} \right) \sin(2\pi x) + e^{-x^2} \left(\frac{d}{dx} \sin(2\pi x) \right) \\ &= \left(-2xe^{-x^2} \right) \sin(2\pi x) + e^{-x^2} \left(\frac{d}{dx} \sin(2\pi x) \right) \end{aligned}$$

Let's continue with $\frac{d}{dx} \sin(2\pi x)$.

Example: $f(x) = e^{-x^2} \sin(2\pi x)$

Chain rule: $\sin(2\pi x)$

As we identify before, $\sin(2\pi x) = \sin(x) \circ 2\pi x$, therefore,

$$\begin{aligned}\frac{d}{dx} \sin(2\pi x) &= \frac{d}{dx} \sin(x) \Big|_{x=2\pi x} \left(\frac{d}{dx} 2\pi x \right) \\ &= \cos(x) \Big|_{x=2\pi x} \cdot 2\pi \\ &= \cos(2\pi x) \cdot 2\pi \\ &= 2\pi \cos(2\pi x)\end{aligned}$$

Example: $f(x) = e^{-x^2} \sin(2\pi x)$

Replace

Let's replace the previous result into the solution,

$$\begin{aligned} f'(x) &= \left(-2xe^{-x^2}\right) \sin(2\pi x) + e^{-x^2} \left(\frac{d}{dx} \sin(2\pi x)\right) \\ &= \left(-2xe^{-x^2}\right) \sin(2\pi x) + e^{-x^2} (2\pi \cos(2\pi x)) \end{aligned}$$

Now we can move on to simplify the expression.

Example: $f(x) = e^{-x^2} \sin(2\pi x)$

Simplify

$$\begin{aligned} f'(x) &= \left(-2xe^{-x^2}\right) \sin(2\pi x) + e^{-x^2} (2\pi \cos(2\pi x)) \\ &= 2e^{-x^2} [-x \sin(2\pi x) + \pi \cos(2\pi x)] \\ &= 2e^{-x^2} [\pi \cos(2\pi x) - x \sin(2\pi x)] \end{aligned}$$

$$f'(x) = 2e^{-x^2} [\pi \cos(2\pi x) - x \sin(2\pi x)]$$

Class Activity

Compute the derivative of the following functions using the rules.

$$f(x) = e^{-\frac{x^2}{4}} \sin\left(\frac{\pi}{2}x + \pi\right), \quad g(x) = \sin^2(x)$$

Answers

$$f'(x) = \frac{1}{2}e^{-x^2/4} \left[\pi \cos\left(\frac{\pi}{2}x + \pi\right) - x \sin\left(\frac{\pi}{2}x + \pi\right) \right]$$

$$g'(x) = 2 \sin(x) \cos(x)$$